

Task 5

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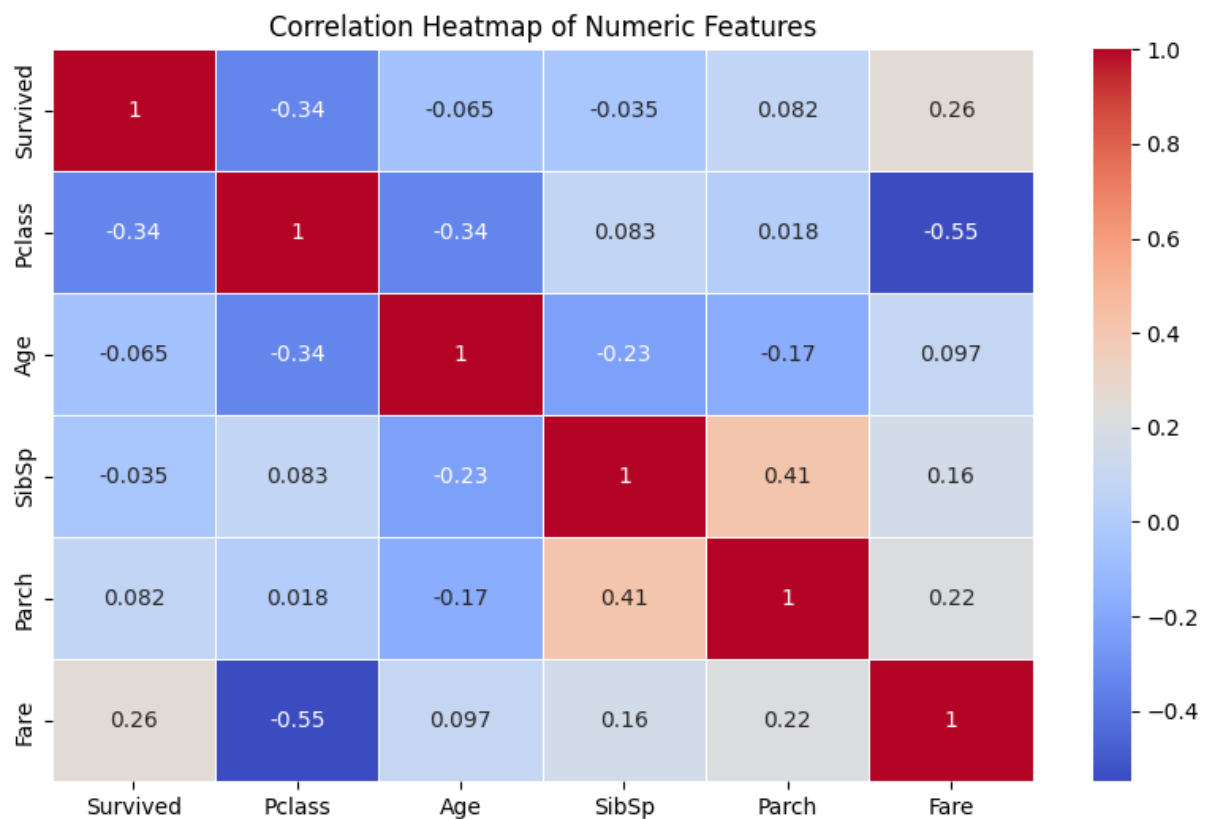
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Correlation Heatmap:

Pclass and Survived show moderate negative correlation (lower class → lower survival).

Fare has weak positive correlation with Survived.

SibSp and Parch are positively correlated (related to family onboard).



Pclass and **Fare** are negatively correlated (-0.55), meaning passengers in higher classes paid higher fares.

Survival has a moderate **positive correlation** with **Fare** (0.26) and **negative correlation** with **Pclass** (-0.34), indicating that passengers in higher classes with higher fares were more likely to survive.

Correlation of **Age** with Survival is very weak (-0.065), meaning age alone is not a strong predictor.

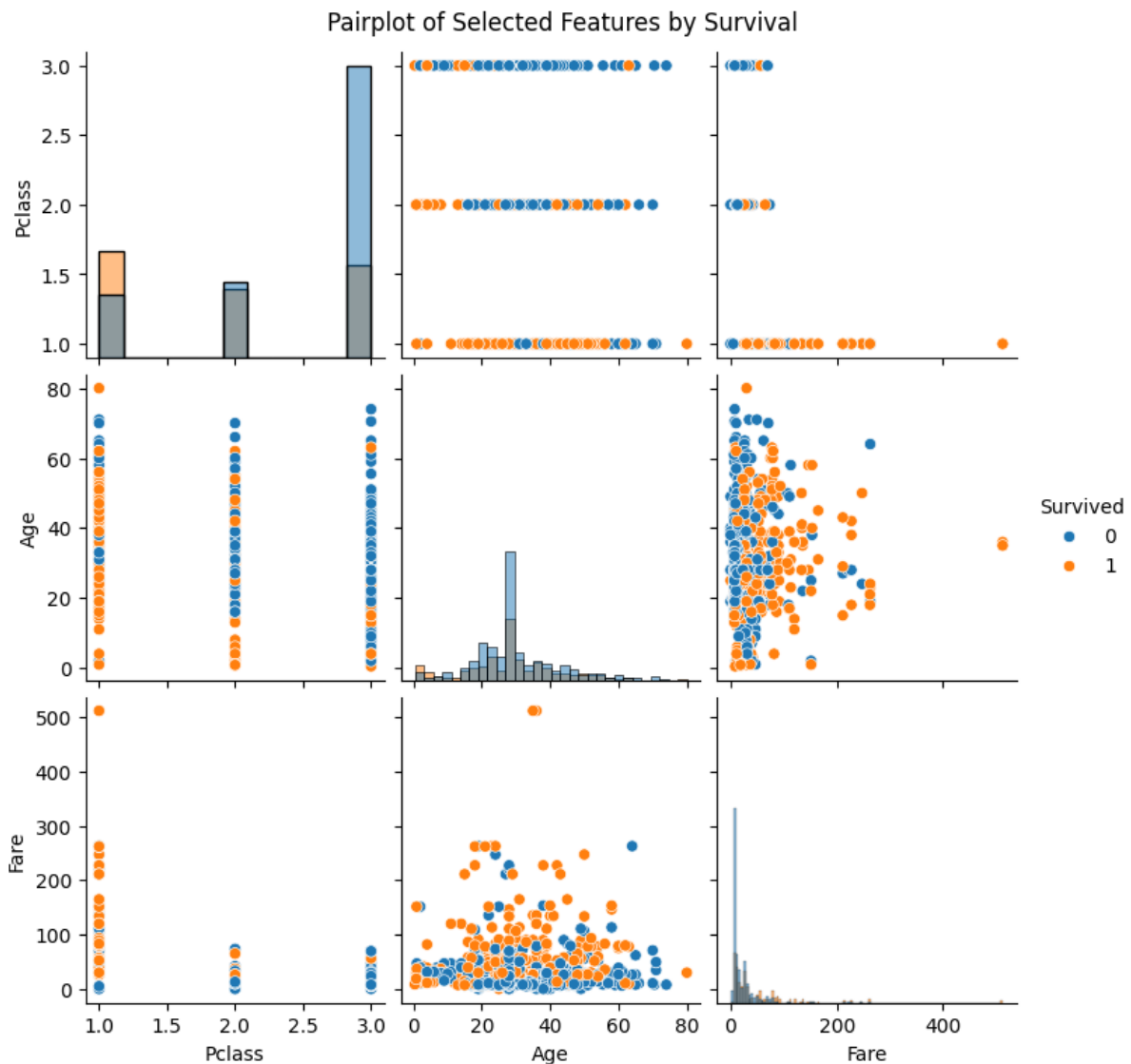
SibSp and **Parch** show some positive correlation with each other (0.41) and moderate correlation with **Survival** (around 0.08–0.16), suggesting family presence had a mild effect.

Pairplot:

Most 1st class passengers had higher fares and a higher survival rate.

Younger passengers (lower Age) had slightly better survival odds.

Clear separation between low vs. high Fare with survival tendency.



Survivors (orange) were mostly in **Pclass 1 and 2**.

Non-survivors (blue) dominate **Pclass 3**, which supports the class-survival link.

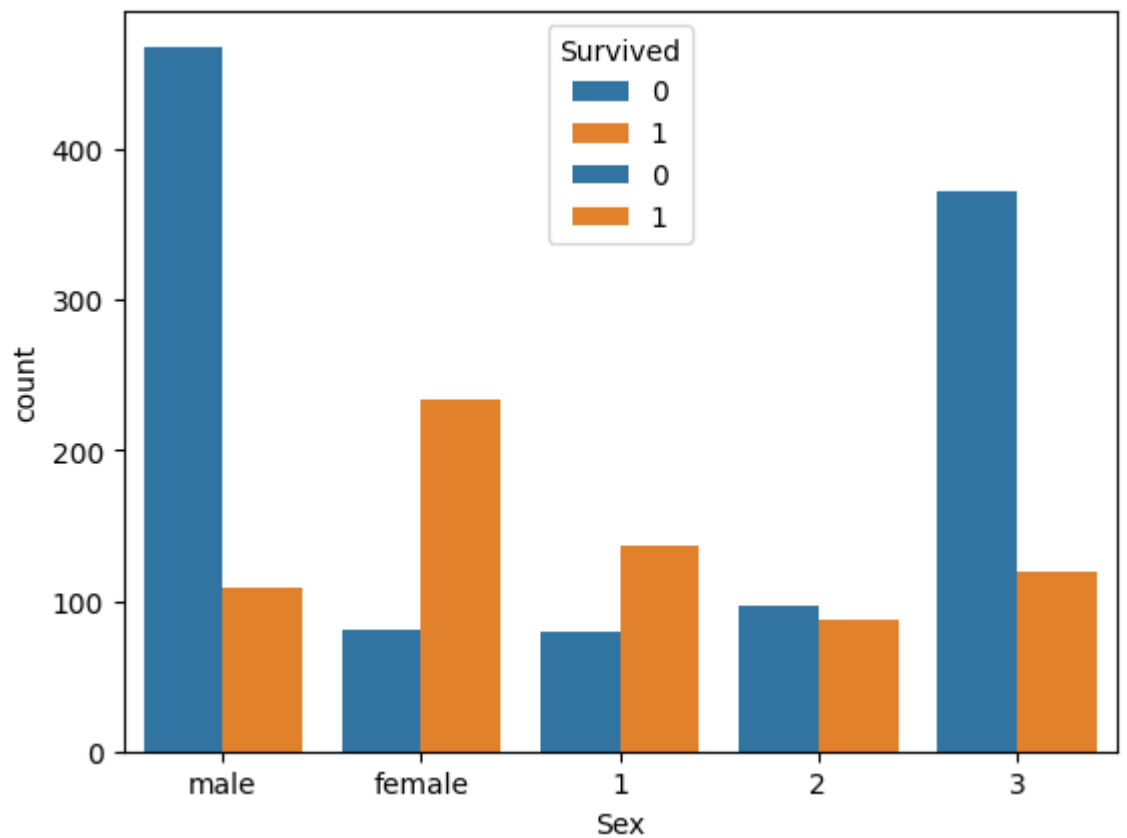
Younger passengers are spread across survival outcomes, but higher **fare** seems to **correlate with survival**, especially at high price outliers.

Count Plot of Sex and Pclass by Survival:

Females had higher survival rates, while most males did not survive.

Pclass 1 passengers had better survival rates compared to **Pclass 3**.

This confirms the influence of **gender** and **class** on survival.

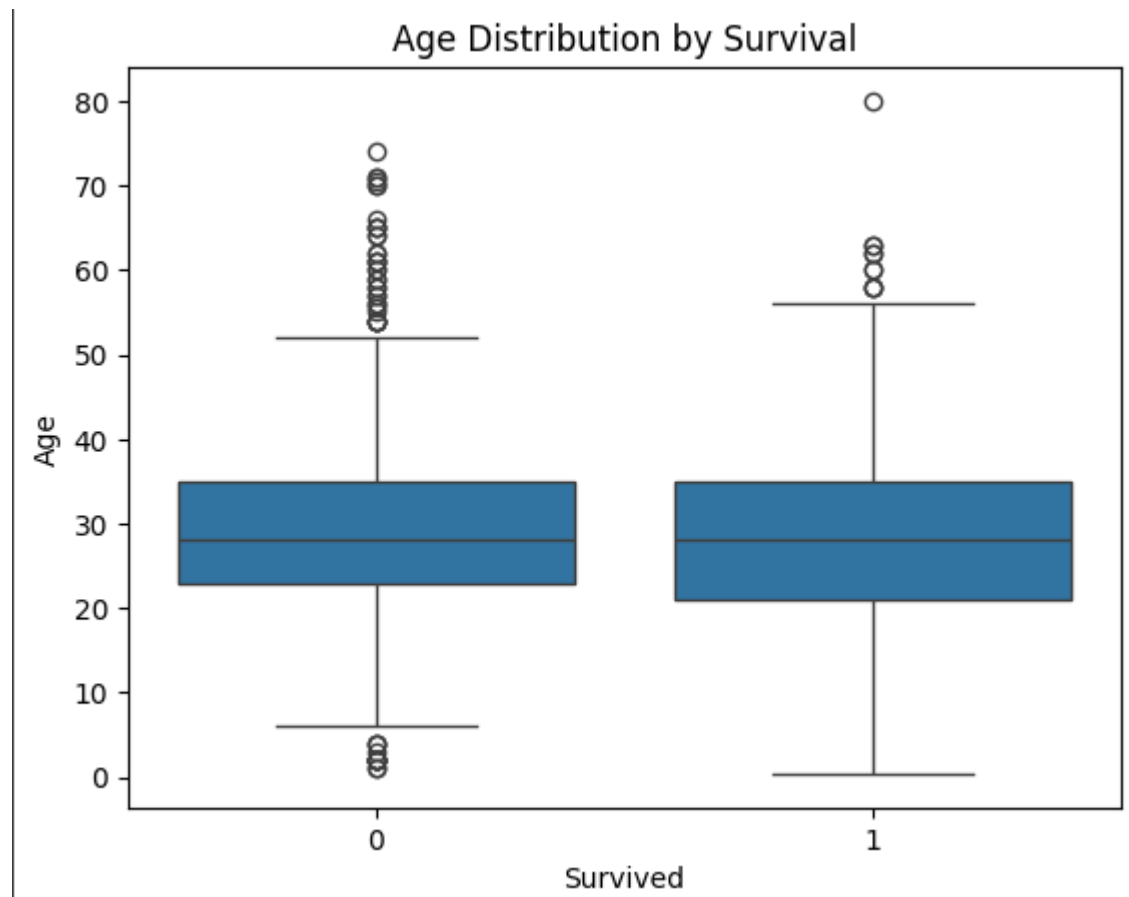


Boxplot of Age by Survival:

Median age is similar for both survivors and non-survivors.

However, **more outliers (older passengers)** are visible among **non-survivors**.

Indicates age alone is **not a decisive factor** but might interact with other variables like class or sex.



Scatterplot of Fare vs. Age Colored by Survival:

Survivors (orange) cluster more in **higher fare** areas.

Most **non-survivors** are in **low fare zones**, often below 100.

This strengthens the case that **wealthier passengers had better survival odds**.

